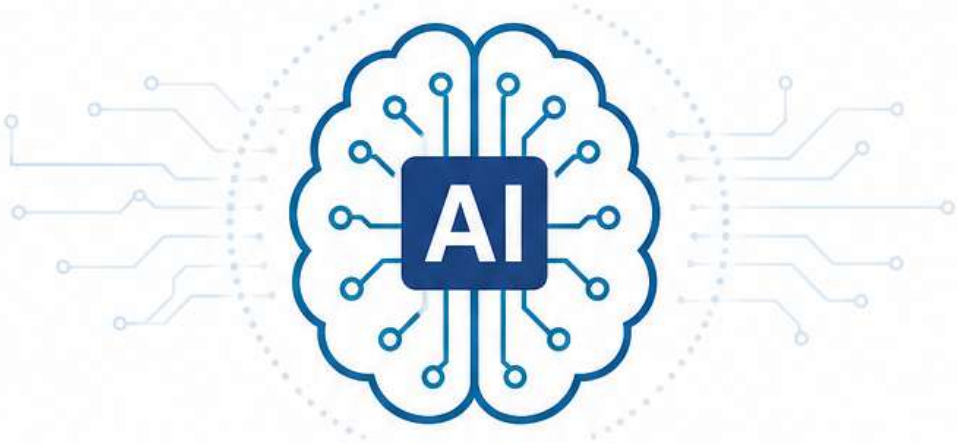
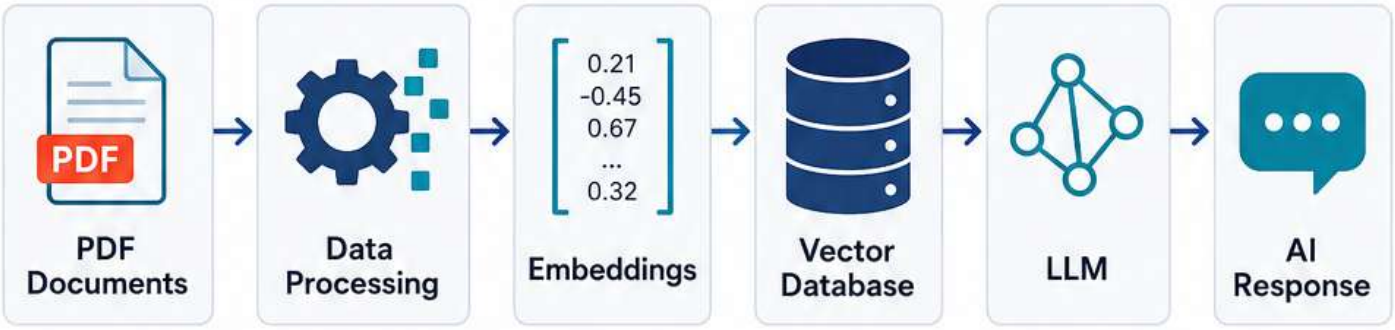
2026 EDITION

SCALABLE RAG ARCHITECTURE

From Raw Documents to AI Answers





Learn how modern AI systems use **Retrieval-Augmented Generation (RAG)** to deliver accurate answers.

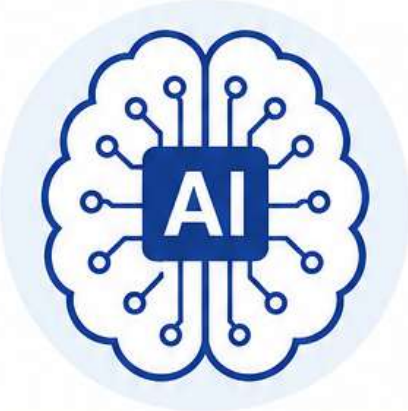
Neural DeepLearn Academy
Deep Learning | Generative AI | Real-World Impact

01

WHAT IS RAG?

Retrieval-Augmented Generation (RAG) helps AI answer questions using your own knowledge sources instead of relying only on its training data.

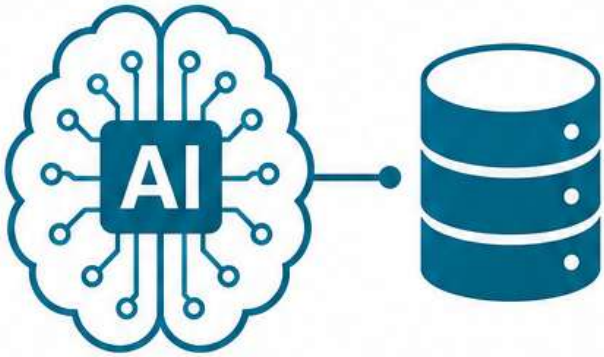
TRADITIONAL LLM



- Limited Knowledge
- Hallucinations
- Outdated Information

VS

RAG SYSTEM

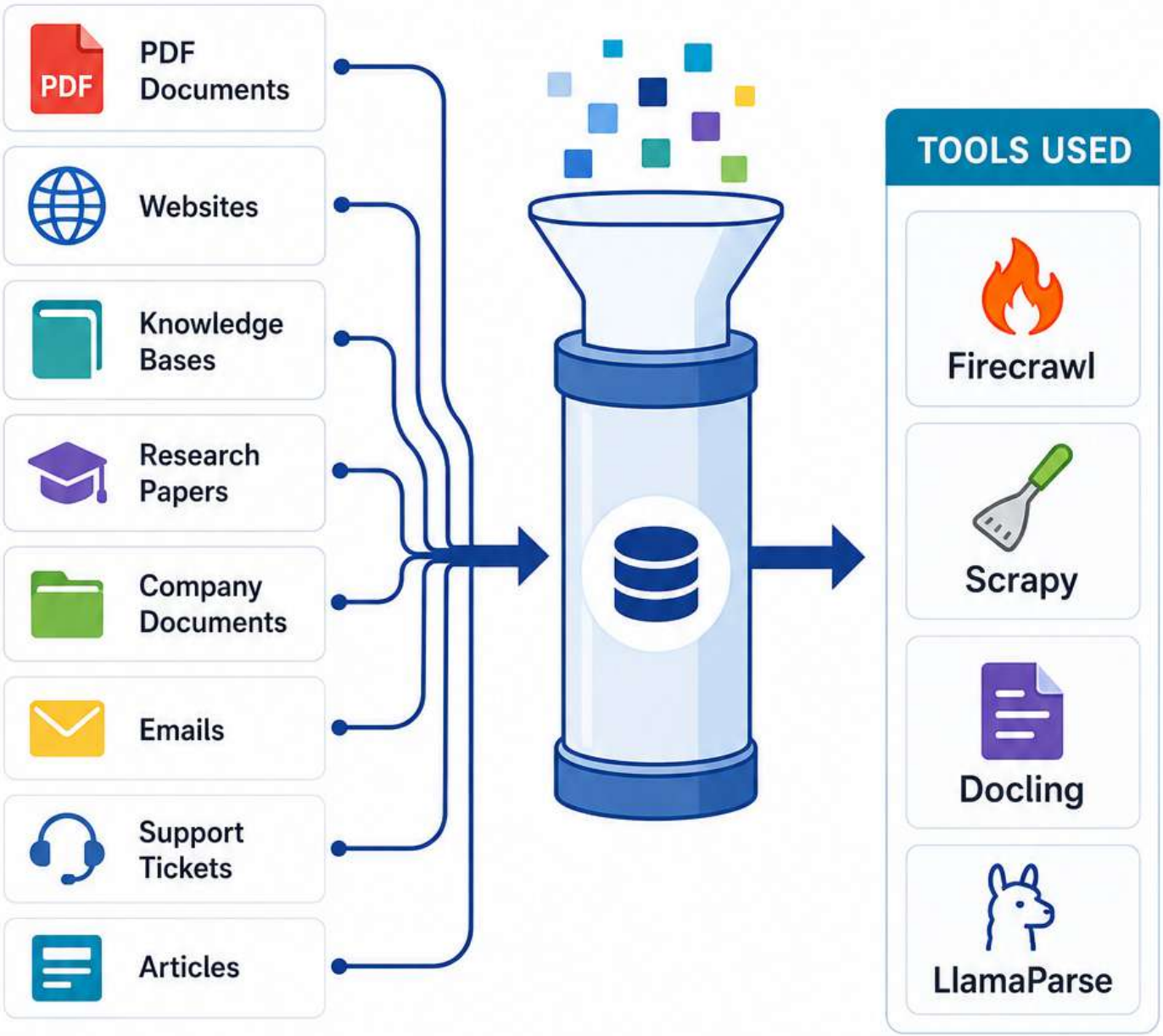


- Accurate Answers
- Fresh Information
- Source Grounded

RAG = Retrieval + Context + Generation

STEP 1
COLLECT DATA

Every RAG system starts with data.



Good Data = Better AI Answers

STEP 2
PROCESS DOCUMENTS

Transform raw documents into structured knowledge.

- RAW DOCUMENT
- CLEANING
- PARSING
- CHUNKING
- METADATA EXTRACTION
- READY FOR EMBEDDINGS

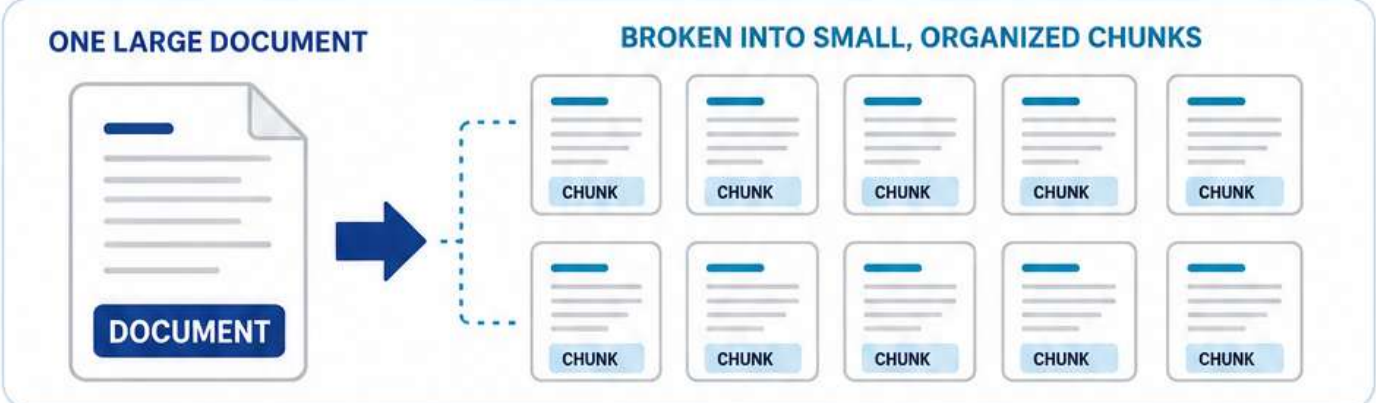
WHY PROCESSING MATTERS

Better Search
Structured content makes information easier to find.

Higher Accuracy
Clean and relevant content leads to more accurate answers.

Lower Hallucination
Grounded responses reduce the risk of AI making things up.

Faster Retrieval
Well-processed data enables quicker and more efficient retrieval.



RAG PIPELINE

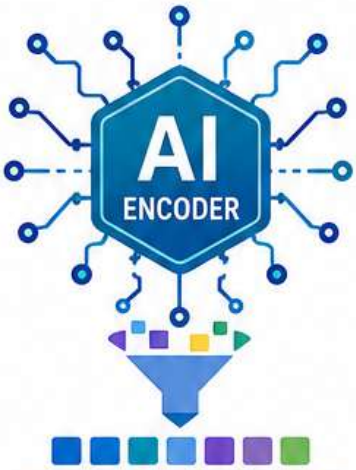
STEP 3
GENERATE EMBEDDINGS

Convert text into vectors that machines can understand.

TEXT CHUNK



EMBEDDING MODEL

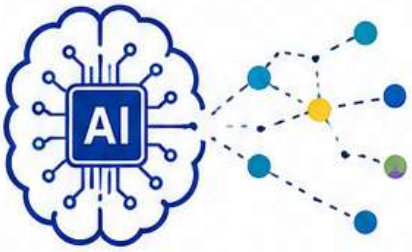


Text → Meaning

VECTOR REPRESENTATION



Machine-readable meaning



WHAT HAPPENS?

Embedding models convert human language into numerical representations that preserve semantic meaning.



POPULAR EMBEDDING MODELS

- OpenAI Embeddings
- Voyage AI
- Nomic Embed
- BGE Models
- E5 Models



EMBEDDINGS ENABLE

SEMANTIC SEARCH
INSTEAD OF
KEYWORD SEARCH

KEYWORD SEARCH
= Exact words

SEMANTIC SEARCH
= Meaning-based retrieval

STEP 4
STORE KNOWLEDGE

Store embeddings for fast semantic retrieval.

TEXT CHUNKS

Raw content split into chunks

EMBEDDINGS

Convert chunks into vectors

VECTOR DATABASE

Store vectors for semantic search



POPULAR VECTOR DATABASES

- Pinecone
- Milvus
- Weaviate
- Qdrant
- Redis Vector Search
- pgvector

TRADITIONAL DATABASES

Store Records

VS

VECTOR DATABASES

Store Meaning

VECTOR DATABASES ENABLE

SIMILARITY SEARCH

INSTEAD OF

KEYWORD MATCHING

STEP 5

USER ASKS A QUESTION

Everything starts with a user query.

Explain Agentic AI in simple terms Search

USER QUESTION

The user asks a question in natural language.

EMBEDDING CONVERSION

The question is converted into a vector representation.

SIMILARITY SEARCH

The system searches the vector database for the most relevant information.

GOOD QUESTIONS LEAD TO BETTER RETRIEVAL.

EXAMPLES

- Summarize this document
- Explain machine learning
- Compare GPT and Claude
- Analyze customer feedback

STEP 6

RETRIEVE RELEVANT CONTEXT

Find the most relevant information.

QUESTION

"Explain Agentic AI in simple terms"

SIMILARITY SEARCH

Search the vector database for similar embeddings.

TOP MATCHING CHUNKS

Return the most relevant chunks.

VECTOR DATABASE



Relevant Chunks (High Similarity)

Irrelevant Chunks (Low Similarity)

WHAT RETRIEVAL DOES

Finds relevant knowledge
Retrieves the most useful information for the query.

Reduces hallucinations
Grounds the model in real, reliable sources.

Improves answer quality
Better context leads to more accurate and complete answers.

Uses company-specific information
Leverages your private data and domain knowledge.

THE QUALITY OF RETRIEVAL OFTEN MATTERS MORE THAN THE CHOICE OF LLM.

STEP 7
GENERATE RESPONSE

Combine context with reasoning.

1 USER QUESTION

"Explain Agentic AI in simple terms."

+

2 RETRIEVED CONTEXT

3 LARGE LANGUAGE MODEL

4 FINAL RESPONSE

POPULAR LLMS

OPEN SOURCE

- Llama 4
- DeepSeek R1
- Gemma 3

PROPRIETARY

- GPT-4o
- Gemini 2.5 Pro
- Claude 4 Sonnet
- Command R+

THE LLM ANSWERS USING RETRIEVED KNOWLEDGE, NOT MEMORY ALONE.

STEP 8

EVALUATE PERFORMANCE

Measure before you deploy.

RAG EVALUATION DASHBOARD

All Projects

ACCURACY
How correct is the answer?
92 / 100

FAITHFULNESS
Is the answer grounded in the context?
93 / 100

RELEVANCE
How relevant is the retrieved context?
88 / 100

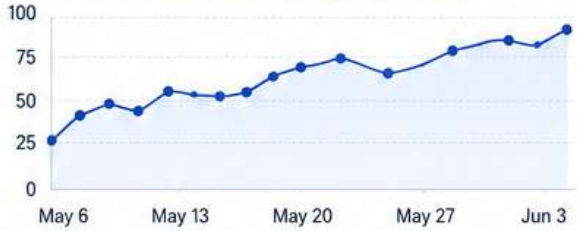
LATENCY
How fast is the system response?
78 / 100
Avg: 1.32s

ANSWER QUALITY
Overall quality of the final answer.
90 / 100

OVERALL SCORE



PERFORMANCE TREND (LAST 30 DAYS)



EVALUATION TOOLS

LangSmith
Observe, test and evaluate LLM apps.

Arize Phoenix
LLM observability and evaluation platform.

DeepEval
Simple and reliable LLM evaluation framework.

Ragas
Evaluate RAG pipelines with custom metrics.

Langfuse
Open-source LLM engineering platform.

YOU CANNOT IMPROVE WHAT YOU DO NOT MEASURE.

COMPLETE RAG WORKFLOW

End-to-End Architecture



1 DATA SOURCES

Collect data from multiple sources (docs, web, emails, databases, etc.).



2 DOCUMENT PROCESSING

Extract text and clean content for processing.



3 CHUNKING

Split documents into smaller, manageable chunks.



4 EMBEDDINGS

Convert chunks into vectors using an embedding model.



5 VECTOR DATABASE

Store embeddings in a vector database for fast similarity search.



6 USER QUERY

User asks a question in natural language.



7 RETRIEVAL

Retrieve the most relevant chunks from the vector database.



8 LLM

Combine retrieved context with the query and generate a response.



9 ANSWER

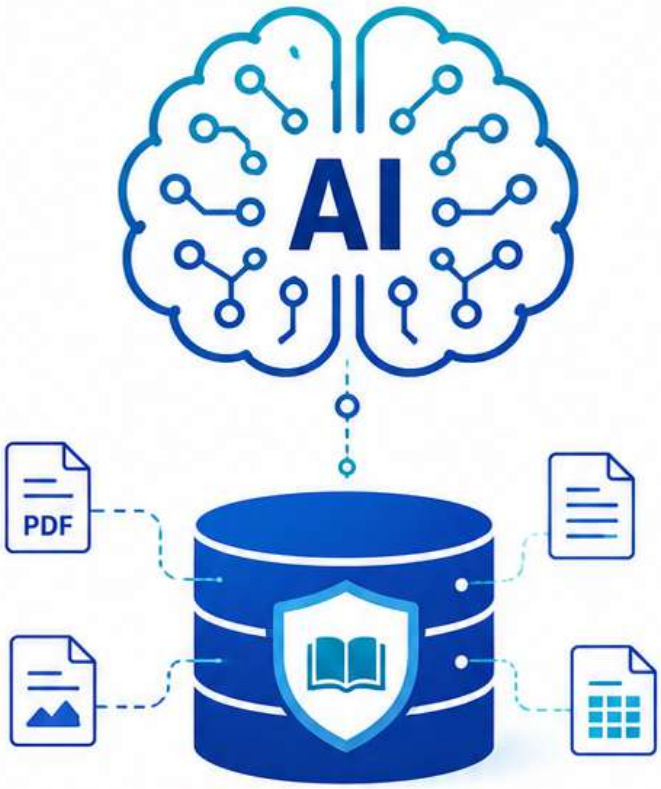
Return the final answer to the user.



DATA + RETRIEVAL + LLM

= RELIABLE AI

KEY TAKEAWAYS



- Uses your own data
- Reduces hallucinations
- Improves answer accuracy
- Delivers explainable answers
- Scales across enterprise knowledge
- Powers modern AI assistants

BUILD AI THAT KNOWS YOUR DATA

Follow Neural DeepLearn Academy
for AI, Machine Learning,
RAG and Agentic AI insights.